Institutional Setting

The 1970s saw a period of increased interest in basic income experiments across North America. In the US, social programs that provided subsidized food, childcare, and healthcare were offered to reduce poverty in the postwar period. In Canada, this lead to a guaranteed annual income project in various communities across Manitoba dubbed the Mincome experiment. The main difference between this experiment and others are the presence of a saturation site — a town in which everyone is eligible to participate in the experiment. For this trial, Dauphin, Manitoba was chosen as the saturation site. This allows for different effects to be studied such as network effects within the community. Additionally, there is an element of game theory included as everyone knows others' basic income level.

From 1974 to 1979, the mincome experiment ran in Manitoba, Canada. In Winnipeg, low-income households were selected to participate in the experiment while in Dauphin, everyone was eligible to receive payments. This not only provided insights into saturation sites, but also includes an urban city and a more rural town to compare. The policy implemented was not a flat payment, but a Negative Income Tax. This means that based size, families would receive a minimum income, which was reduced for every dollar earned through work. There were different treatment levels on the tax rate of total income.

Although no official final reports were released from the government, years later federal funding allowed the University of Manitoba to release the database of results for Mincome, letting individuals create their own research. Some of these analyses showed a small impact on labour markets, with small decreases in working hours (Hum & Simpson, 1993). Others studies have focused on health and social benefits.

Empirical Design/Strategy and Data

Labour supply is generally measured by hours worked. We can therefore define productivity as output over hours worked. Using worker productivity, φ_{it}, as the explanatory variable, let the model be defined as:

$$\varphi_{it} = \frac{y_{it}}{h_{it}} = x_{it}\beta + \alpha_i + \eta_t + \varepsilon_{it},\ i = 1\ ,\ldots,n;\ t = 1\ ,\ldots,T$$

Where h_{it} is hours worked per week for household i at time t. Then y_{it} is the output from that same household at the same time period. x_{it} is the vector of observable attributes about the household (our main variables of interest) while α_i are unobservable individual effects such as personal motivation or stress. Finally, η_t represents time-fixed effects and ε_{it} is the error term.

In order to derive causal impacts from this data, we must make the assumption that given an individual's observable attributes, the effect of any unobservable effects must be zero. i.e. $\mathbb{E}[\alpha_i \,|\, x_i] = 0,\ \forall i$. This assumption allows the model to estimate based on fixed individual effects which gives us the following:

$$\varphi_{it} - \varphi_i = \frac{y_{it}}{h_{it}} - \frac{y_i}{h_i} = [x_{it} - x_i]\beta + [\eta_t - \eta] + [\varepsilon_{it} - \varepsilon_i]$$

Note that α_i cancels out here due to it not depending on t. In this model φ_i, y_i, h_i, x_i and η are the means for their respective variable over all time periods. Let Δ be our DiD estimator, with $D_i = 1$ if household is treated, and $G_i = 1$ if the household is a racialized household. Then the difference-in-differences model is:

$$\Delta = \beta_0 + \beta_1 D_i + \beta_2 G_i + \beta_3 (D_i \times G_i) + \beta_4 X_{it} + \alpha_i + \eta_t + \varepsilon_{it}$$

where β_1 is the average treatment effect on labour productivity and β_3 is the differential treatment effect for racialized households.

For this model the main variables to considered are the response variable, weekly hours worked and weekly income. For x_{it} we want to include general household variables such as double-head or single-headed household, family size, highest education received, site (Winnipeg, Dauphin, rural), and treatment (how much money household was given). For studying the effects of this program on racialized peoples the variables ethic group and date entered Canada will be particularly useful. Looking at the individual's net worth is also a characteristic to consider as those with wealth may not react to a guaranteed income program as strongly. These variables will allow for a comparison of families new to Canada versus those with well-established roots in the country. Additionally, looking at a breakdown of the results by race could help to determine the differing effects on racialized households.

In addition to the assumption listed above that lets us construct the model ($\mathbb{E}[\alpha_i | x_i] = 0, \forall i$), we must also assume strict exogeneity: $\mathbb{E}[\varepsilon_{it} | x_{it}, \alpha_i] = 0$, this conditional allows for the separation of treatment effects from other effects. A Hausman-Wu test, comparing the coefficients of a fixed effects model versus a random effects model, would potentially be useful to determine whether this assumption is violated.

The parallel trend assumption is also key to derive causal effects. This is the foundation of the model as it allows for the comparison of two groups, only changing the treatment variable to isolate the treatment effect. Assuming these two groups would act

the same, ceteris paribus, ensures the model coefficients are identifiable. To test for the parallel trend assumption, a pre-trend test will be performed verifying the claim that before the intervention, these groups had similar working hours and output before treatment.

Another assumption made is the stable unit treatment value assumption which assumes the treatment group does not interfere with the control group's effect, i.e. there are no spillover effects. For our purpose, the treatment of one household must not affect another's labour supply. This could happen if say two people working are at the same employer, the treatment reduces working hours, then the control must increase their hours to cover. Because of the presence of the saturation site, we are able to compare saturation versus a non-saturated area. If the wage or labour effects vary greatly from the other sites, then there may be general equilibrium effects that violate SUTVA.

There are some key parts of this experiment that may cause bias, mainly the lack of randomization of treatment and the temporary nature of the program. Firstly, although the treatment selection in the non-saturation sites (Winnipeg and rural) were randomly assigned, there have been some criticism with the method used for assignment. Additionally, only households below a certain income threshold were selected to participate in the experiment. This lower income is likely correlated with a lower weekly working hours, meaning the treatment effect may be larger than results, underestimating the true effect, the bias is positive.

The second source of potential bias is related to the temporary nature of the program. Because participants expect the program to end, they may not change their behaviour to change as it would if the program were permanent. For example, a household that is treated, continues to work as normal because they know they will need to work af-

ter the experiment ends. This would lead to an underestimated effect from treatment, meaning negative bias.

Data Sources & Analysis Sample

The Mincome data used in this analysis is publicly available through the University of Manitoba's data verse on "borealis", a repository for Canadian data. The data verse contains various datasets and codebooks that provide meaningful insight into the variables and what data was collected. Not only can the data be accessed on this site, but also past papers, and working papers using this data can be found in the data verse as well.

Over the course of 5 years (1974-1979), survey data was collected resulting in 750 total observations, each of which contained over 500 variables. Households were only retained if there was baseline information and if the last two years of payment information was complete. The data is divided into sub-datasets containing information on family structure, employment information of the adults in the household, personal information/characteristics of individuals, wealth, etc.

Despite the researchers considering different environments, urban, small town, and rural settings, these sites may not be representative of other urban Canadian cities like Toronto or Vancouver, or remote groups such as indigenous communities. Due to the age of the experiment, the demographic data has drastically changed since the 1970s with different immigrant composition and more women in the workforce, these factors are not representative of modern Canada. The focus on low-income households must also be considered, as results may not generalize to a universal program.

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   FAMSIZE        TOTFAMINC74       HHHRWRK        RACIALHH         AVGPROD
 Min.   : 1.000   Min.   :    0   Min.   :   0   Min.   :0.0000   Min.   :  0.000
 1st Qu.: 2.000   1st Qu.: 4522   1st Qu.: 600   1st Qu.:0.0000   1st Qu.:  7.705
 Median : 3.000   Median : 6825   Median : 775   Median :0.0000   Median : 10.284
 Mean   : 3.302   Mean   : 7178   Mean   : 693   Mean   :0.0479   Mean   : 11.862
 3rd Qu.: 5.000   3rd Qu.: 9203   3rd Qu.: 800   3rd Qu.:0.0000   3rd Qu.: 14.294
 Max.   :12.000   Max.   :25911   Max.   :1399   Max.   :1.0000   Max.   :209.000
 NA's   :250      NA's   :271     NA's   :588                     NA's   :598
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Table 1. Summary statistics of key variables. From left to right, family size, total family income in 1974, household hours worked, does household have a racial minority, average productivity (total family income / household hours worked).

For the purposes of this analysis, racial minority is defined as someone from the following ethnic groups: Filipino, Chinese, African, West Indian, Black/African descent, Japanese, South American. Below is a breakdown of counts from the racial household variable, where 1 represents a racialized household.

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  0   1
954  48
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